

Darius Pétermann

Researcher in the field of applied machine learning and signal processing for audio & music

 Scholar

 Website

 Email

 LinkedIn

 GitHub

1 EDUCATION

Ph.D. in Intelligent Systems Engineering - Computer Science

May 2025

Luddy School of Informatics, Computing, and Engineering, Indiana University (GPA: 4.0 / 4.0)

Bloomington, IN

- **Selected Coursework:** Machine Learning for Signal Processing, Deep Learning, Computer Vision, Applied Machine Learning
- Research Group: Signals and AI Group in Engineering (SAIGE) 
- Thesis: “Efficient Native Neural Sub-band Coding through Residual Feature Representation within Hyper-Autoencoded Reconstruction Propagation Networks”
- Advisor: Prof. Minje Kim 

M.Sc. in Sound and Music Computing

Sep 2020

Dept. of Information and Communication Technologies, Universitat Pompeu-Fabra (GPA: 9.5 / 10.0)

Barcelona, Spain

- **Selected Coursework:** Music Information Retrieval, System Design, Audio Signal Processing, ML for Audio, Research Methods, Reinforcement Learning
- Thesis: “SATB Voice Segregation for Monaural Recordings” 
- Advisor: Dr. Pritish Chandna 

B.M. in Music Synthesis - Audio Engineering

May 2016

Electronic Production & Design Dept, Berklee College of Music (GPA: 3.8 / 4.0)

Boston, MA

- **Selected Coursework:** Digital Signal Processing, Physical Computing, Audio Programming in C, Principles of Audio Electronics, Music Acoustics
- Thesis: “A Deep Look at Spectral Synthesis Techniques Through csConvolve” 
- Advisor: Dr. Richard Boulanger 

2 POSITIONS HELD

Amazon

March 2025 – Present

Applied Scientist, L5

Bellevue, WA

- Conducting research in the Audio Foundation Team, focusing on advancing multimodal foundation models
- Developing large-scale models for audio and music understanding, reasoning, and representation learning

Netflix

Jun 2024 – Nov 2024

Machine Learning Research Intern

Los Gatos, CA

- Supervised by Dr. Mahdi Kalayeh 
- Working on multimodal (audio-visual) generative AI problems

Google Research

May 2023 – Oct 2023

Student Researcher

Cambridge, MA

- Supervised by Dr. Hakan Erdogan , Dr. John Herhsey , and Dr. Scott Wisdom 

- Working on next-level unsupervised audio source separation problems

Mitsubishi Electric Research Labs (MERL)

Summers in 2021 - 2022

Research Intern

Cambridge, MA

- Received \$15'000 of gift-money from MERL to work on the “Cocktail Fork Problem”
- Hosted by Dr. Gordon Wichern  and Dr. Jonathan Le Roux 
- Derived and implemented new models and optimization methods for audio analysis with applications to source separation in challenging multi-source environments and using advanced machine learning techniques

Signals and AI Group in Engineering (SAIGE), Indiana University

Jan 2021 - Present

Research Assistant

Bloomington, IN

- Supervised by Prof. Minje Kim 
- Conducting research pertaining to neural audio coding and audio source separation problems

Senseable Intelligence Group, Massachusetts Institute of Technology (MIT)

Nov 2020 - Apr 2021

Technical Lab Assistant

Cambridge, MA

- Contractor for Senseable Intelligence Group, McGovern Institute for Brain Research, led by Dr. Satrajit Gosh 

Apple Inc.

Jun 2016 - Jul 2019

Content Engineer

Cupertino, CA

- Software engineer for Apple’s pro Audio & Music Apps (LogicPro, GarageBand) 
- Designed real-time MIDI processing systems in C++ for Apple’s virtual musical instruments

Electronic Production & Design Dept, Berklee College of Music

Sep 2015 - May 2016

Programming Tutor

Boston, MA

- Tutored and mentored EPD students for technical classes: “Audio Programming in C”, “Digital Signal Processing”, “Csound”, “Max/MSP”

3 PUBLICATIONS

Preprints and Submissions

- [R04] *Discrete Audio Tokens: More Than a Survey!*, Submitted to Transactions on Machine Learning Research (TMLR), Under Review, 2025. 
- [R03] *Seeing Sound: Assembling Sounds from Visuals for Audio-to-Image Generation*, arXiv, 2025. 
- [R02] *Topic: Benchmarking discrete representations for audio and speech tasks*, Submitted to NeurIPS, Under Review, 2025.
- [R01] *Topic: High-efficiency neural audio coding using hierarchical and residual multi-band quantization*, Submitted to NeurIPS, Under Review, 2025.

International Conference Papers (Peer reviewed)

- [C08] J. Shi, H.-j. Shim, J. Tian, *et al.*, “VERSA: A versatile evaluation toolkit for speech, audio, and music,” in *2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics – System Demonstration Track*, 2025. [Online]. Available: <https://openreview.net/forum?id=zU0hmbnyQm>.  
- [C07] D. Petermann and M. Kim, “Hyperbolic distance-based speech separation,” in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2024, (to appear).  

- [C06] **D. Petermann**, I. Jang, and M. Kim, “Native multi-band audio coding within hyper-autoencoded reconstruction propagation networks,” in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2023, pp. 1–5.    
- [C05] **D. Petermann**, G. Wichern, A. Subramanian, and J. L. Roux, “Hyperbolic audio source separation,” in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2023, pp. 1–5.      **[Best Student Paper Award & Top 3% Papers]**
- [C04] **D. Petermann** and M. Kim, “SpaIn-Net: Spatially-informed stereophonic music source separation,” in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2022, pp. 106–110.    
- [C03] **D. Petermann**, G. Wichern, Z.-Q. Wang, and J. L. Roux, “The cocktail fork problem: Three-stem audio separation for real-world soundtracks,” in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2022, pp. 526–530.    
- [C02] **D. Petermann**, S. Beack, and M. Kim, “Harp-net: Hyper-autoencoded reconstruction propagation for scalable neural audio coding,” in *Proc. of the IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2021, pp. 316–320.    
- [C01] **D. Petermann**, P. Chandna, H. Cuesta, J. Bonada, and E. Gomez, “Deep learning based source separation applied to choir ensembles,” in *Proc. of the International Society for Music Information Retrieval Conference (ISMIR)*, 2020, pp. 733–739.    

Journal Articles

- [J02] **D. Petermann**, G. Wichern, A. S. Subramanian, Z.-Q. Wang, and J. L. Roux, “Tackling the cocktail fork problem for separation and transcription of real-world soundtracks,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 2592–2605, 2023.   
- [J01] P. Chandna, H. Cuesta, **D. Petermann**, and E. Gómez, “A deep-learning based framework for source separation, analysis, and synthesis of choral ensembles,” *Frontiers in Signal Processing*, vol. 2, 2022. 

Patents

- [P02] S. K. Beack, W. Lim, I. Jang, *et al.*, *Audio signal encoding/decoding methods and apparatus for performing the same*, US Patent App. 63/420 405, 2023.
- [P01] **D. Petermann**, G. Wichern, A. Subramanian, and J. L. Roux, *Audio source separation using hyperbolic embeddings*, US Patent App. 18/191 417, 2023.

4 HONORS AND AWARDS

Best Student Paper Award & Top 3% Papers Recognition <i>ICASSP 2023</i>	Jun 2023
• For the paper entitled “Hyperbolic Audio Source Separation” 	
Outstanding Reviewer Award Recognition <i>ICASSP 2023</i>	Jun 2023
• Awarded to the top 5% of the reviewers (220/4445)	
“Excellent” Grade with Honor, Music Cognition & Perception <i>Universitat Pompeu-Fabra</i>	Sep 2020

Magna Cum Laude Honor

Berklee College of Music

May 2016

Dean's List

Berklee College of Music

May 2016 - May 2016

- Appeared on Berklee Dean's List for 7 semesters out of 8

BT Production Award & Scholarship

Berklee College of Music

April 2015

- Award and scholarship from the Electronic Production & Design Dept

5 TUTORIALS, OTHER TALKS & POSTERS

Invited talk for the class “Machine Learning for Musicians” 

Dept. of Electronic Production & Design (EPD), Berklee College of Music

Nov 2021

Boston, MA

Poster Presentations (Non-Archival) on “Hyperbolic Audio Source Separation”

Luddy AI Center Open House, Indiana University

Mar 2023

Bloomington, IN

6 CERTIFICATIONS

Machine Learning

Stanford - Coursera

Oct 2019

Audio Signal Processing for Music Applications

Universitat Pompeu-Fabra - Coursera

Jul 2018

7 TECHNICAL SKILLS

Programming Languages: Python, C++, Objective-C, C, Lua, Javascript, L^AT_EX

Deep Learning (over 6 years of experience): PyTorch, PyTorch Lightning, TensorFlow, JAX

Audio Signal Processing (over 10 years of experience): JUCE, MATLAB, Csound, MAX/MSP, PureData

8 REFEREES

Dr. Minje Kim  

Former Principal Investigator of the Signals and AI Group in Engineering (SAIGE) at Indiana University.
Associate professor of Computer Science at University of Illinois Urbana-Champaign (UIUC)

Dr. Xavier Serra  

Director of the Music Technology Group at Universitat Pompeu-Fabra, Professor in the Dept. of
Information and Communication Technologies

Dr. Richard Boulanger  

Professor in the Electronic Production & Design Dept. at Berklee College of Music